

History of Agentic AI.

Generated by AutoResearch Agent • April 29, 2026

History of Agentic AI

Foundational Concepts

Agency in AI fundamentally requires:

- **Perception** of environment through sensors
- **Goal representation** and objectives
- **Decision-making** to select actions
- **Action execution** to change the environment

Early agency centered on explicit symbolic reasoning about state transitions toward desired goals.

Classical Era (1970s-1990s)

STRIPS and symbolic planning systems featured:

- Explicit world state representations using formal logic
- Complete information assumptions
- Deterministic action effects
- Offline planning before execution
- Strong on puzzles, weak on real-world complexity

Machine Learning Transition (1990s-2010s)

The shift to machine learning introduced:

- **Learned behaviors** replacing hand-coded rules
- **Adaptive policies** improving through experience
- **Reinforcement learning** enabling autonomous goal-seeking through trial-and-error
- Systems like AlphaGo demonstrating superhuman planning via neural networks

This addressed symbolic AI's brittleness and inflexibility.

Modern LLM-Based Agents (2022+)

Recent breakthroughs enabling current implementations:

- **Emergent reasoning** from scaled transformer models
- **Instruction-tuning and RLHF** creating reliable tool-users
- **In-context learning** enabling dynamic task adaptation
- **Function-calling APIs** standardizing tool integration

Key Differences from Earlier Architectures

Aspect	Classical	Modern
Knowledge	Hand-coded	Learned from pretraining
Action selection	Rigid state machines	Natural language composition
Generalization	Task-specific	Cross-domain
Planning	Explicit offline	Implicit in-context

Modern agentic AI leverages learned world models and demonstrates flexible task adaptation without explicit programming, representing a fundamental shift from deterministic logical systems to probabilistic, learned approaches.